COMS30035, Machine learning: k-means and mixtures of Gaussians

James Cussens

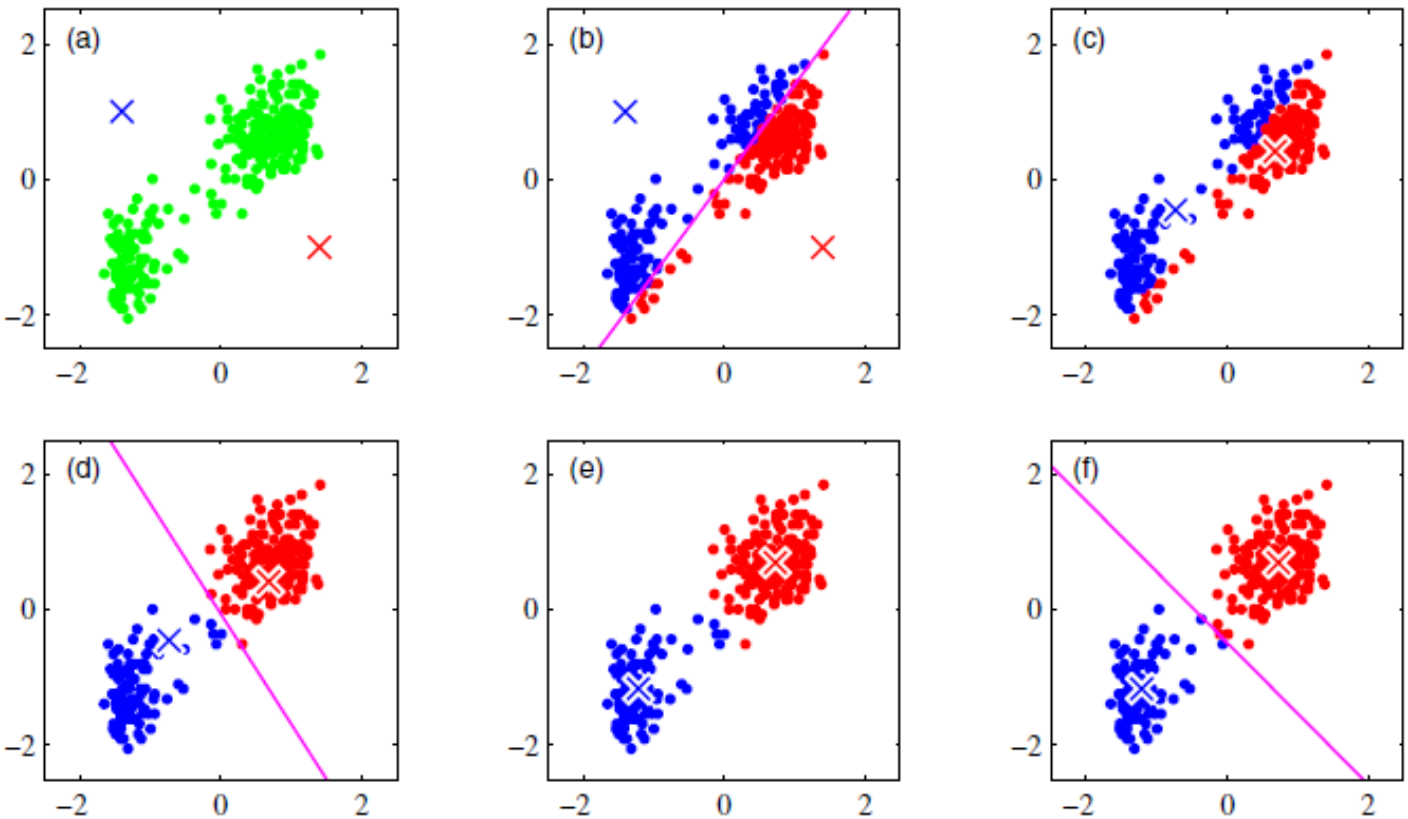
School of Computer Science
University of Bristol

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Clustering

- Supervised vs Unsupervised learning.
- **Two methods:**
  - k-means: simple distance-based clustering.
  - Gaussian Mixture Models (GMMs): probabilistic soft clustering.

k-means for clustering



k-means optimisation

- $r_{nk} = 1$ if datapoint $\mathbf{x}_n$ is assigned to cluster k.
- $\boldsymbol{\mu}_k$ is the mean of cluster k.

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_j ||\mathbf{x}_n - \boldsymbol{\mu}_j||^2 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

$$\boldsymbol{\mu}_k = \frac{\sum_n r_{nk}\mathbf{x}_n}{\sum_n r_{nk}} \tag{2}$$

k-means objective function

- We want each point $\mathbf{x}_n$ to be close to its assigned cluster mean $\boldsymbol{\mu}_k$.
- Objective: minimize total squared distance to cluster centres.

$$J = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$$

- Optimize by alternating:
  1. Initialize K centres.
  2. Repeat until convergence (cluster assignments stop changing).
     - **E-step:** Assign each $\mathbf{x}_n$ to its nearest $\boldsymbol{\mu}_k$.
     - **M-step:** Update each $\boldsymbol{\mu}_k$ as the mean of its assigned points.

Pros and cons of k-means

Advantages

- Simple and fast: scales linearly with the number of data points N, clusters K, and dimensions D.
- Easy to implement and interpret — each cluster is summarized by its mean $\boldsymbol{\mu}_k$.
- Works well for roughly spherical and well-separated clusters.
- Often a good first choice for exploratory data analysis or initialization (e.g., for EM in GMMs).

Disadvantages

- Requires pre-specifying the number of clusters K.
- Assumes clusters are isotropic (spherical) and of similar size.
- Sensitive to initialization — can get stuck in local minima.
- Sensitive to outliers and noise (means can be pulled by extreme points).
- Produces *hard* assignments only — no uncertainty in cluster membership.

Gaussian mixture distribution

A Gaussian mixture model (GMM) assumes that data is generated from a mixture of K Gaussian components: [Bis06, §9.2]

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{3}$$

We can associate the *mixing coefficients* π_k with a K-dimensional latent variable z where:

$$z_k \in \{0, 1\}, \quad \sum_k z_k = 1, \quad p(z_k = 1) = \pi_k \tag{4}$$

Then, the marginal likelihood can be written as:

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{5}$$

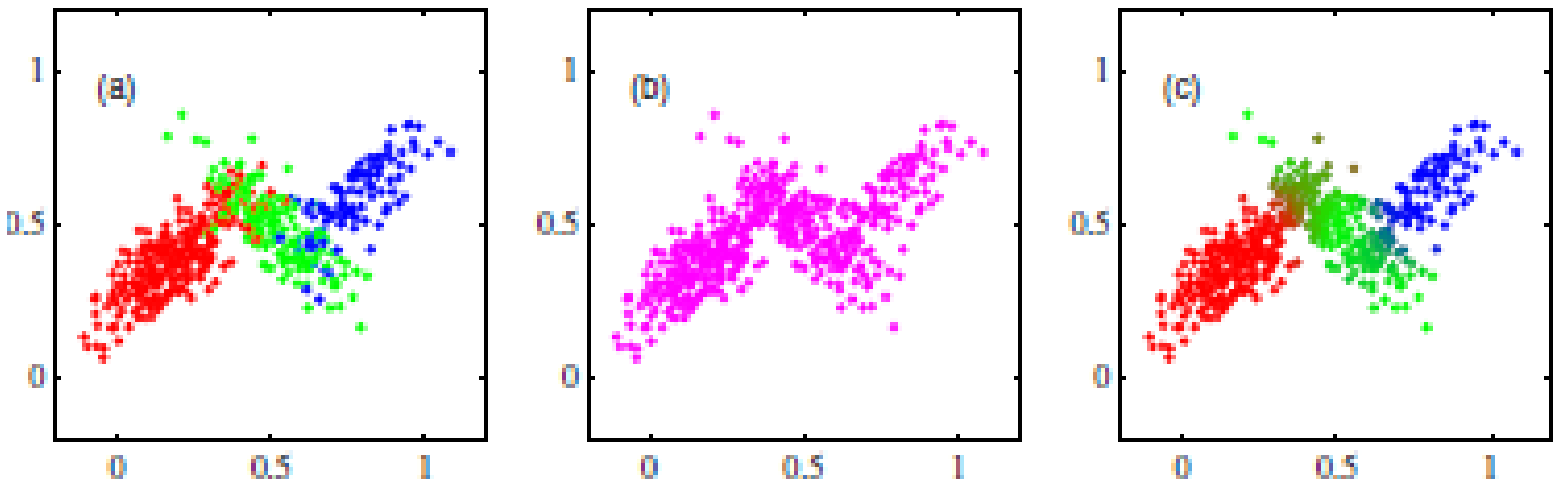
Responsibility and sampling

- Now we have a full joint distribution $p(\mathbf{x}, \mathbf{z})$ we can define the *responsibility* that component k has for 'explaining' observation $\mathbf{x}$

$$\gamma(z_k) = p(z_k = 1|\mathbf{x}) \tag{6}$$

- To sample from a Gaussian mixture just use ancestral sampling: sample from $p(\mathbf{z})$, and then from $p(\mathbf{x}|\mathbf{z})$.

Soft clustering with Gaussian mixtures



More clustering with Gaussian mixtures

- If we want we can put restrictions on the covariance matrices of the Gaussians in the mixture.
  - Spherical: $\sum_k = \sigma_k^2 I$
  - Diagonal: $\sum_k = \text{diag}(\sigma_1^2, \sigma_2^2, ..., \sigma_K^2)$
  - Full: No constraint; captures correlations between features.
- Let's have a look at [Mur22, p.729]

(Soft) clustering by MLE of a Gaussian mixture

- Given data $\mathbf{X}$ (and a fixed number K of component Gaussians) we can use MLE to estimate a particular Gaussian mixture distribution.
- Here's the log-likelihood [Bis06, 433]:

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\} \tag{7}$$

- Number of problems:
  1. Possible singularities
  2. Symmetry/nonidentifiability
  3. No closed form for the MLE
- Queue the EM algorithm ...

Comparing k-means and GMMs

- k-means: hard assignments, spherical clusters.
- GMM: soft assignments, elliptical clusters.
- GMM generalizes k-means when $\Sigma_k = \sigma^2 I$ and π_k are equal.

Reading

- Bishop §9.1 (you can skip §9.1.1 but it's an interesting read).
- Bishop §9.2 up to §9.2.1
- Murphy §21.3 up to §21.3.3

Problems and quizzes

- Bishop Exercise 9.1
- Bishop Exercise 9.3
- Quizzes:
  - Week 5: k-means and Mixtures of Gaussians


Christopher M. Bishop.
Pattern Recognition and Machine Learning.
Springer, 2006.


Kevin P. Murphy.
Probabilistic Machine Learning: An introduction.
MIT Press, 2022.